

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	CH.POORNA TEJA	Reg. No.:	192472149
Section / Batch:	2024	Date:	22/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q47. Application: Data Mining Pattern Extraction System A data mining system processes large datasets using the recurrence $T(n) = 4T(n/4) + n \log n$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze scalability for large data mining tasks. Interpret efficiency using asymptotic notation.

Q48. Application: Cloud Resource Allocation System A cloud computing system allocates resources using the recurrence $T(n) = 6T(n/3) + n$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how performance scales with increasing cloud workloads. Interpret efficiency using asymptotic notation. Generate a concept map that makes attractive and understanding way to copy paste in. Word document

Code of Conduct

I Ch.poorna teja-192472149 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student POORNA

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature:	Signature:	Signature:
Date:	Date:	Date:

Q47.

A Data Mining Pattern Extraction System analyzes very large datasets to discover hidden patterns, trends, and relationships. To process data efficiently, the dataset is divided into 4 equal subproblems, each solved recursively.

The recurrence relation is:

$$T(n) = 4T(n/4) + n \log n$$

Using the Master Theorem:

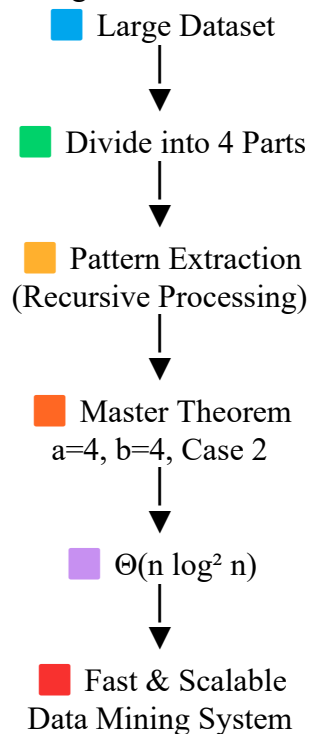
- $a = 4$
- $b = 4$
- $f(n) = n \log n$
- $n^{\log_4 4} = n$

Since $f(n) = \Theta(n \log n)$, it satisfies Master Theorem Case 2.

Time Complexity:

$$\boxed{\Theta(n \log^2 n)}$$

This means the algorithm scales efficiently for large datasets and is suitable for modern data mining applications.



Q48.

A Cloud Resource Allocation System distributes computing resources such as CPU, memory, and storage among

multiple users. The workload is divided into 6 recursive subproblems, each of size $n/3$. The recurrence relation is:

$$T(n) = 6T(n/3) + n$$

Using the Master Theorem:

- $a = 6$
- $b = 3$
- $f(n) = n$
- $n^{\log_3 6} \approx n^{1.63}$

Since $f(n) = O(n^{1.63-\epsilon})$, it satisfies Master Theorem Case 1.

Time Complexity:

$$\Theta(n^{1.63})$$

This shows the system scales well as cloud workloads increase, making it efficient for resource management.

